

MARL for Cooperative Payload Transport & Collision Avoidance

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Multi-agent deep reinforcement learning (MAPPO + QMIX, CTDE) for a team of 2–4 differential-drive rovers that must cooperatively drag a shared rigid payload to a goal zone through static and dynamic obstacles — with a classical potential-field baseline for comparison and a real-time PyGame demo.

Quickstart

```
# 1. install dependencies (Python 3.11)
pip install -r requirements.txt

# 2. train (each command resumes safely if interrupted – just re-run it)
python src/train.py --algo mappo --iters 40 --curriculum --max-stage 3 --run-name mappo_n3_cur --resume
python src/train.py --algo qmix --frames 140000 --qmix-levels 3 --curriculum --max-stage 3 --run-name qmix_n3_s0 --resume

# 3. formal benchmark (30 episodes, writes runs/eval_*.json/.csv)
python src/evaluate.py --ckpt runs/qmix_n3_s0/ckpt/best.pt --episodes 30
python src/evaluate.py --ckpt runs/mappo_n3_cur/ckpt/best.pt --episodes 30
python src/evaluate.py --baseline --episodes 30
python src/evaluate.py --random --episodes 30

# 4. real-time visual demo
python src/demo.py --ckpt runs/qmix_n3_s0/ckpt/best.pt # trained QMIX
python src/demo.py --ckpt runs/mappo_n3_cur/ckpt/best.pt # trained MAPPO
python src/demo.py --baseline # classical controller
python src/demo.py --random # untrained, for contrast

# 5. regenerate every report figure
python src/plots.py
python src/trajectories.py

# 6. rebuild the submission PDFs (REPORT.pdf incl. figures appendix)
python src/make_pdf.py
```

Demo controls: **SPACE** pause · **R** new scenario · **+/-** speed · **ESC** quit.

Repository layout

Path	Purpose
src/env.py	Cooperative payload-transport environment: shared-rig physics, LiDAR obs, CTDE global state, cooperative team reward
src/mappo.py	MAPPO — shared Beta-policy actor + centralized critic, GAE, clipped PPO
src/qmix.py	QMIX — per-agent Q nets, monotonic mixing network, Double-Q, n-step replay
src/baseline.py	APF classical controller (stations on forward ring + tangential escape + stuck back-off)
src/train.py	Training entry point with curriculum staging, eval, CSV logs, checkpoints

Path	Purpose
src/evaluate.py	Formal 30-episode benchmark → metrics JSON/CSV
src/demo.py	Real-time PyGame visualization
src/plots.py, src/trajectories.py	Report figures
REPORT.md	Full written report (methodology, results, discussion)
runs/	Training logs, eval JSONs, checkpoints (best.pt, final.pt)
plots/	Generated figures

Mapping to the proposal slides

Slide	Where it lives in this repo
4 — System architecture & AI/ML flow	env.py (sim) → mappo.py/qmix.py (CTDE training) → evaluate.py/demo.py (deployment-style inference)
5 — Simulation & training environment	PayloadTransportEnv: 2–4 rovers, 12×12 m arena, 5 static + 1 dynamic obstacle, constrained starts, domain randomization via reseeding
6 — MARL engine (CTDE)	Local obs (9-beam LiDAR, rel. goal/payload/teammates) for actors; global state (all poses, payload vel, rig stretches) for the centralized critic/mixer; cooperative reward = cohesion + stability – collisions
7 — Execution, demo & metrics	demo.py real-time run; metrics: success rate, collisions, path efficiency, rig stability (evaluate.py)
8 — Industrial relevance	Discussed in REPORT.md (warehouse AMR fleets, precision agriculture, cooperative lifting)
9 — Literature	The four verified IEEE-format references cited in REPORT.md
10 — Roadmap & deliverables	Survey→algorithm selection (Phase 1) → environment (Phase 2) → training + ablations (Phase 3) → real-time demo + metrics report (Phase 4) — all complete

Training results at a glance (30-episode formal eval, full task)

Policy	Success	Collisions/ep (mean)	Steps (successes)	Notes
Random	0%	1.6	—	sanity floor
MAPPO (curriculum)	~27%	21.5 (0 on every success)	432	careful, collision-averse
QMIX (9-action, best ckpt)	50%	75.5	224	fast, exploits contacts
APF baseline*	100%	2.8	81	*privileged ground-truth sensing

See REPORT.md for full methodology, ablations and discussion.